Perpetual American Put

A verified finite-difference obstacle solver

Coursework team

Project 2 technical draft

The decision to exercise creates an obstacle

- Exercise gives $(K - S)^+$ immediately, continuation may be more valuable.
- The stationary one-dimensional problem is $\min(-\mathcal{L}V, V - \psi) = 0$.
- We seek both the value and the exercise threshold.

Monotone differences preserve the obstacle structure

- Upwind drift produces a tridiagonal M-matrix.
- Policy iteration alternates exercise and continuation rows.
- The stopping test measures complementarity residual.
- The contact node uses price-gap tolerance 10^{-7}.

Exact prices converge at roughly first order

- Stage 1 exact threshold is $500/7$.
- Four meshes reduce maximum price error from 0.05847 to 0.00736.
- Last contact node on the finest mesh is 71.4375.
- Nodal threshold errors need not decrease monotonically.

Stage 1 benchmark results

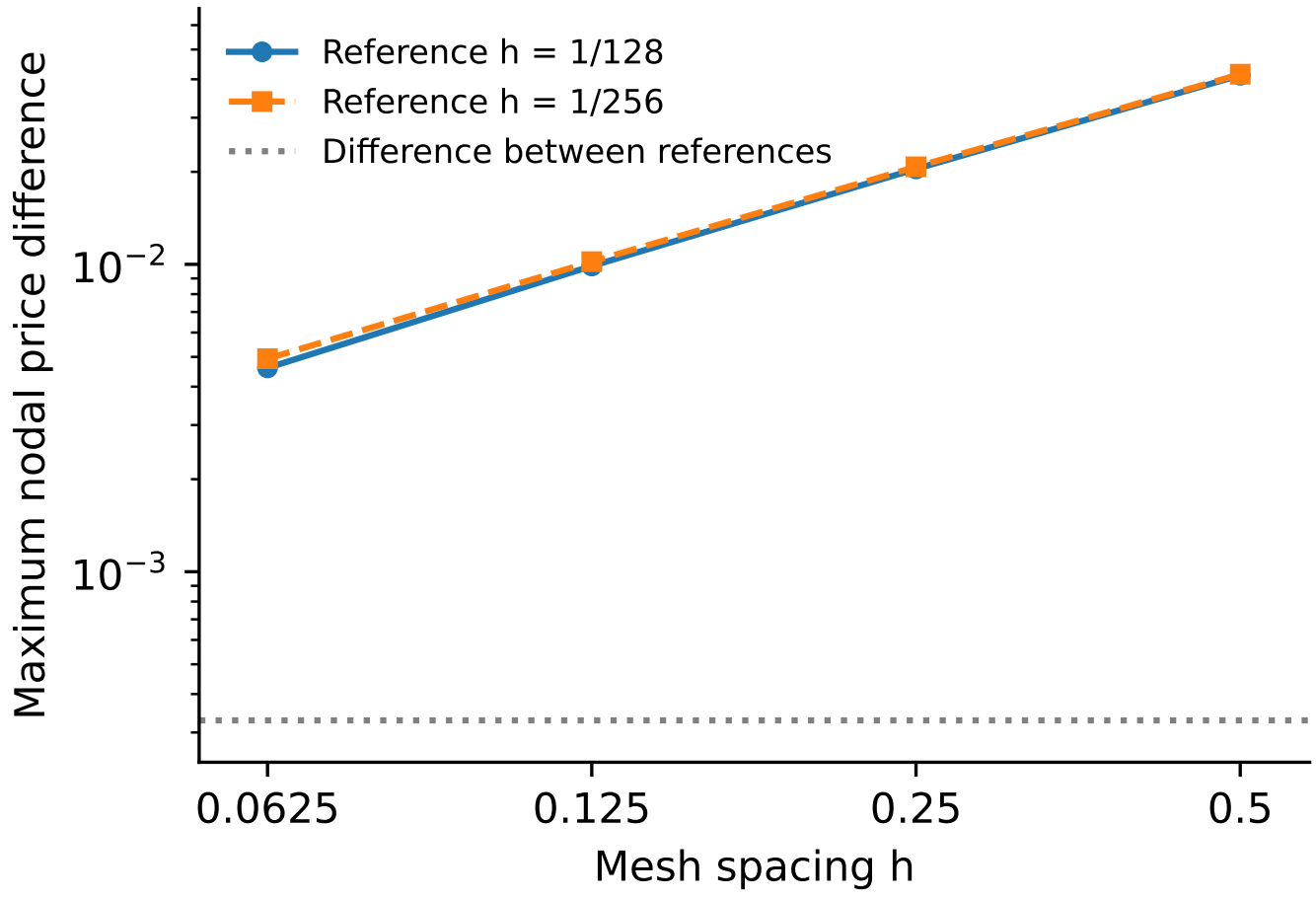
Exact threshold $= 500/7 \approx 71.42857$

h	Max price error	Contact node
0.5	0.058473	71.5000
0.25	0.029348	71.2500
0.125	0.014735	71.3750
0.0625	0.007363	71.4375

Stage 2 converges on a fixed truncated domain

- Local volatility and $q = 0.02$ produce a genuinely numerical case.
- At $L = 600$, finest mesh versus fine-grid difference is 0.004604.
- A second reference changes the price comparison by $3.28 \cdot 10^{-4}$.
- Fine-reference residuals reach a reported floating-point floor.

Stage 2 grid convergence



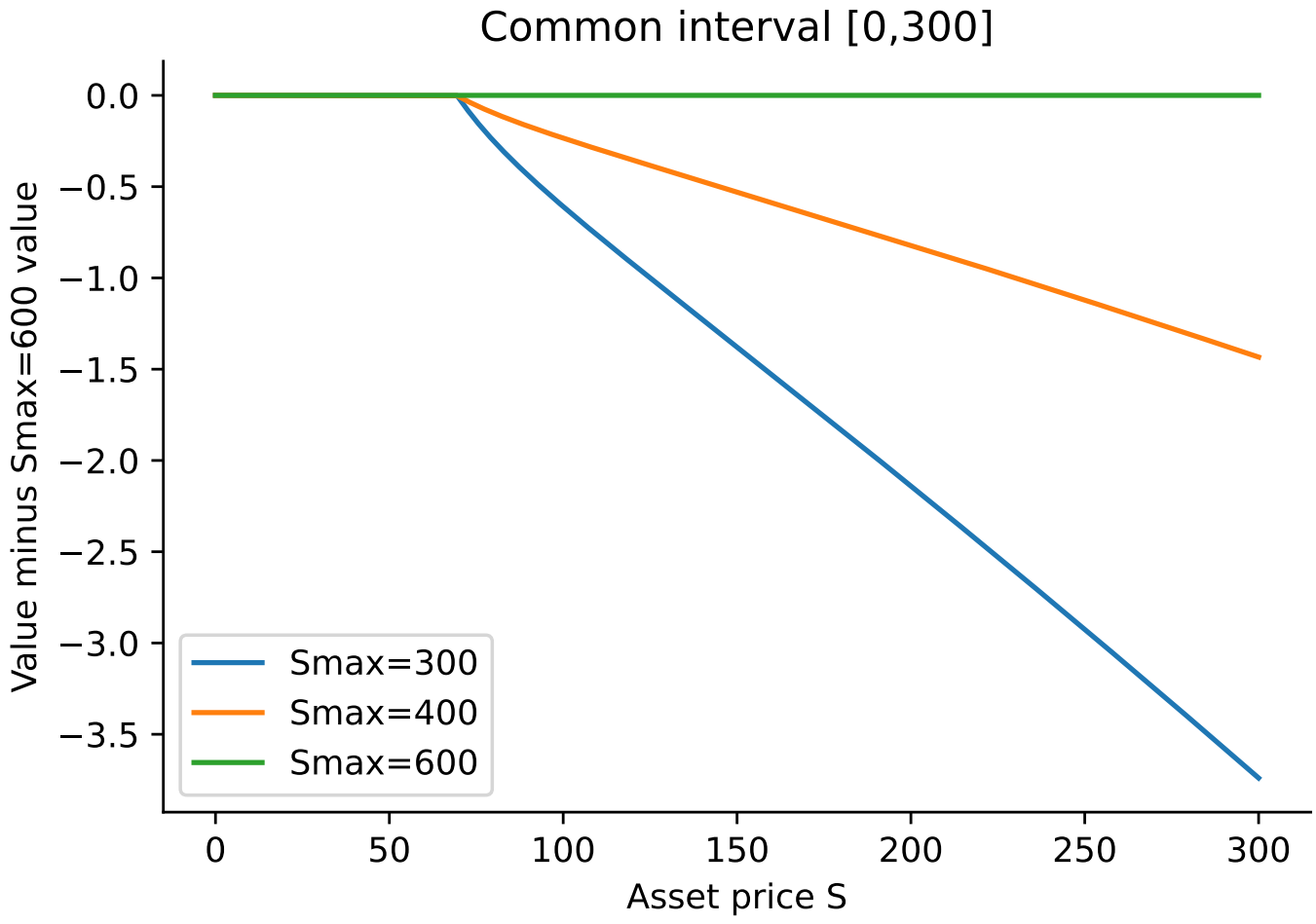
Fixed domain [0, 600]

The outer boundary changes the answer materially

L	numerical $V(100)$	collocation $V(100)$
300	15.4560	15.4512
400	15.8310	15.8260
600	16.0653	16.0601

Domain enlargement and mesh refinement measure different sensitivities.

Stage 2 domain sensitivity



Fixed mesh spacing $h = 0.0625$

Independent checks support the finite-domain result

- Analytic Stage 1 solution checks price and threshold.
- Exhaustive small-grid active sets check policy iteration.
- Shooting and adaptive collocation agree for the same domain.
- Shape, complementarity and tolerance diagnostics are recorded.

Two solvers agree at a common residual target

- Policy iteration and projected SOR use the same LCP and payoff initialisation.
- At $h = 5$ and 2.5, maximum solution differences are below $4 \cdot 10^{-8}$.
- Policy systems use compiled linear algebra, while PSOR uses Python sweeps.
- Timings include the implementation and platform effects.

Solver comparison results

Common residual target 10^{-8}

h	Policy iterations	PSOR sweeps	Max price difference
5	6	10260	3.98×10^{-8}
2.5	12	41073	3.89×10^{-8}

Reproducibility and discussion

- Reproduce with the repository README and inspect saved numerical evidence.
- Grid and tolerance studies alone cannot remove truncation bias.
- Explain stencil signs, complementarity and threshold extraction.
- Complete the student audit, individual contributions and presentation rehearsal.